



Quantifying adoption: A SEM study of quantum software technology in software development

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Abstract

Context Quantum software technologies (QSTs) are emerging as a promising field, offering developers powerful tools to push the boundaries of software innovation. However, a technology that is not adopted remains nothing more than an untapped potential. Despite increasing interest in integrating quantum computing into software development, significant barriers still hinder its widespread adoption. While much of the existing research has focused on technical advancements, the socio-technical factors influencing adoption have been largely overlooked. Understanding these factors could provide managers and engineers with actionable insights to facilitate adoption, as developers' perceptions and attitudes can be shaped and managed more readily than technological advancements.

Objective This study addresses this gap by statistically analyzing the determinants of QST adoption among software professionals.

Method Leveraging established adoption theories (UTAUT2 and Diffusion of Innovation) and incorporating perceived trust, data were collected using validated questionnaires and analyzed with Partial Least Squares Structural Equation Modeling (PLS-SEM).

Results Findings indicate that compatibility, personal innovativeness, and performance expectancy are key drivers of adoption, while financial concerns negatively influence it. Moreover, workplace support and habitual exposure emerged as critical enablers of actual usage, emphasizing the importance of organizational interventions.

Conclusions These insights offer practical guidance for companies developing and maintaining QSTs into their workflows while identifying new research directions, including barriers to adoption, long-term engagement, and financial considerations.

Keywords Quantum software engineering · Socio-technical aspects · Technology adoption · PLS-SEM

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1 Introduction

Quantum technologies are increasingly becoming a tangible reality for both practitioners and researchers. The potential of these technologies to offer novel solutions to previously insurmountable problems, as well as to enhance fields that have stagnated due to the limitations of current technology (Knight 2018; Hoare and Milner 2005), has driven significant investment in this emerging area. As a result, *quantum software technologies (QSTs)*—specifically designed to support software development, such as cloud platforms, quantum simulators, and quantum programming languages—have started to emerge and become available within the contemporary software development landscape. Consequentially, quantum computing, a branch of computer science grounded in quantum theory, has garnered substantial attention in both industry and open-source communities. This has led to the integration of quantum technologies into software products, with the potential to deliver significant benefits to the field, such as the development of higher-quality software and advanced machine learning systems capable of facilitating deeper analysis of large, complex datasets (Zhang and Ni 2020).

Nevertheless, a technology that is not adopted leads to wasted resources and missed opportunities. To drive the adoption of QSTs, various stakeholders—both researchers and private companies, such as IBM and Microsoft—have invested in developing more accessible tools (Javadi-Abhari et al. 2024). These efforts, among other objectives, aim to encourage the widespread use of this innovative and promising technology. Particularly, the research community has established the field of *quantum software engineering (QSE)* (Piattini et al. 2021b, 2020, 2021a; Moguel et al. 2020). This emerging discipline seeks to create new solutions, apply quantum computing to a broader range of challenges, and conduct empirical studies to assess the state of practice, particularly focusing on the technical side.

Nonetheless, software engineering is inherently a socio-technical activity, involving not only the development of software artifacts but also the adoption and integration of emerging technologies into software development workflows (Brooks Jr 1995; Ralph et al. 2016; Hoda 2021). Quantum software development is no exception, and the effectiveness of these technologies depends not only on their technical performance but also on their adoption by developers. Understanding adoption dynamics is therefore a critical SE challenge, influencing how QSTs integrate into software development processes, tools, and team practices. Indeed, by examining these aspects, it becomes possible to improve the current state of practice, leading to a better understanding of the adoption and use of quantum technologies, as well as to the development of guidelines and recommendations aimed at enhancing the development cycle.

Despite the growing importance of this research field, studies specifically addressing the socio-technical dimensions of quantum software development remain entirely absent. However, the research community has begun to advocate for increased attention to these aspects (De Stefano et al. 2024b; Murillo et al. 2025; Shaydulin et al. 2020). In particular, the recent work by Murillo et al. (2025) highlights the need for further research on the management of teams developing quantum software, emphasizing the challenges associated with the adoption of QSTs and the need to facilitate such an integration in the development lifecycle. Additionally, and perhaps even more critically, no effort has been made to investigate one of the main (if not the most important) factors in defining socio-technical systems—the first step that leads to identifying an activity as socio-technical: the adoption

of the technology itself. Specifically, to the best of our knowledge, **there are currently no studies in the domain of QSE that explore—by mean of a statistical investigation—the factors influencing software professionals to adopt QSTs.**

We argue that this lack of investigation represents a missed opportunity. The early stages of an emerging technology's lifecycle are critical for shaping its future trajectory (Hall and Khan 2003; Utterback and Suarez 1995). Although the widespread adoption of QSTs may still be several years away, the technology has already reached a point where tools, frameworks, and programming models are available and accessible to practitioners (Javadi-Abhari et al. 2024). This transitional phase—characterized by limited but feasible adoption—is precisely when understanding adoption dynamics can yield the greatest impact. Insights into developers' expectations, pain points, and early interaction patterns can inform the design of tools, educational resources, and organizational strategies that align with real-world needs. Without this knowledge, adoption barriers may go unrecognized, resulting in wasted investments and poor integration into existing workflows. Conversely, identifying these factors early on enables the creation of targeted interventions, ultimately fostering broader acceptance and more effective use of quantum tools. Furthermore, such understanding provides actionable guidance for shaping developer-facing technologies, thereby accelerating innovation and ensuring workforce readiness for future demands (Rogers 2003; Rogers et al. 2014; Venkatesh et al. 2003; Davis et al. 1989).

To address the aforementioned limitations, this work seeks to take an initial yet comprehensive and systematic step toward investigating the adoption of QSTs in software development teams; we aimed to answer the following research question.

❓ RQ — *What are the individual factors that influence the adoption of Quantum Software Technologies (QSTs) in software development?*

Recognizing that these technologies are still in the innovation phase, as highlighted by the Gartner report (Gartner 2023), and aiming to provide a comprehensive view of the complex adoption phenomenon, we decided to study adoption using more than one established adoption theory. Specifically, we operationalized the factors highlighted into two well-established theories: the *Unified Theory of Acceptance and Use of Technology (UTAUT2)* (Venkatesh et al. 2012) and the *Diffusion of Innovations Theory (DoI)* (Rogers 2003). Moreover, we incorporated the *perceived trust* in companies offering QSTs as it was proven to be important to the adoption of different IT technologies. Data collection was conducted using validated questionnaires from the literature with strong reliability metrics, while the analysis was performed using Partial Least Squares Structural Equation Modeling (PLS-SEM) as a robust statistical technique.

The organization of the paper is as follows: Section 2 reviews the pertinent literature. In Section 3, we outline the theoretical framework that underpins the hypotheses essential for the theory under examination. The research methodology and procedures implemented to address our research question are described in Section 4. Section 5 details the outcomes of the statistical analysis, and Section 6 discusses the study's reflections and implications. Finally, Section 7 concludes the paper with final thoughts.

2 Theoretical Background and Related Work

This section describes the background and related work that is the foundation for our contributions.

2.1 Quantum Software Engineering

Quantum computing, a subfield of computer science, harnesses the principles of quantum mechanics to perform computations (Knight 2018; Hoare and Milner 2005). After this, QSE has emerged as a formal research discipline (Moguel et al. 2020; Zhao 2020; Piattini et al. 2020), with its foundational principles outlined in the *Talavera Manifesto* (Piattini et al. 2020). Building upon this work, Piattini et al. (2021b) have further defined specific research domains within QSE, identifying four key areas: the design of quantum-hybrid systems, the development of testing methodologies, the evaluation of quantum program quality, and the re-engineering of classical-quantum information systems. A central challenge highlighted in their work is the need to bridge the knowledge gap between quantum computer scientists and traditional software engineers, as collaboration between these fields is crucial for the advancement of QSE.

The work by Piattini et al. (2021b) has significantly shaped subsequent research in the field of QSE. In the years that followed, the research community started to focus on various topics, like testing and maintenance techniques (e.g., García de la Barrera et al., 2023; Ali, 2024; Leite Ramalho et al., 2025; Yu and Zhao, 2025), architectural design principles (e.g., Zhao et al., 2024), and the characterization of development processes and associated challenges (e.g., Dwivedi et al., 2024; Zappin et al., 2025). More recently, increasing attention has also been devoted to security and vulnerability concerns in quantum software systems (e.g., Santa Barletta et al., 2025; Mao et al., 2025; Nadim et al., 2025).

Expanding on the work by Piattini et al. (2021b); De Stefano et al. (2024a) conducted a systematic mapping study to assess the current research landscape in QSE. Their study revealed that much of the existing research has focused disproportionately on software testing, highlighting a lack of studies on the factors influencing the adoption of QSTs—an area that remains relatively unexplored. Moreover, contributors (Unitary Fund 2024) and researchers (De Stefano et al. 2024b; Shaydulin et al. 2020) have emphasized the need for greater attention to these aspects, particularly given the interdisciplinary nature of quantum software development.

Recently, Murillo et al. (2025) proposed a research roadmap, identifying key gaps in QSE. Their study outlined several critical challenges that the QSE community should prioritize, including: ensuring quality assurance in quantum systems through improved testing and debugging methodologies (Long and Zhao 2024; Ye et al. 2023); shifting towards a service-oriented approach to quantum software development (Valencia et al. 2021; Romero-Álvarez et al. 2023); exploring model-driven quantum development (Ali and Yue 2020; Gemeinhardt et al. 2021); and deepening research on software engineering processes, particularly in the management of teams developing QST-based systems, which are inherently socio-technical in nature.

This last point is particularly relevant to our work, as it underscores the need for studies that provide structured knowledge to support development teams and managers in integrating QSTs into software engineering workflows. Our study directly contributes to this

research agenda by offering insights into the adoption dynamics of QSTs, ultimately facilitating their integration into software development teams and processes.

2.2 Adoption of Quantum Software Technologies

The adoption of QSTs is an increasingly studied topic in the computer science community.

As highlighted by Bisicchia et al. (2024), one of the main challenges to the adoption of QSTs is that it is primarily confined to a restricted group of experts due to the high complexity of quantum programming and the need for advanced abstractions. The limited availability of high-level development tools means that many developers must work directly with low-level quantum concepts, creating a barrier to broader adoption. Furthermore, systematic investigations into the current practices of quantum programming are sparse, underscoring the need for comprehensive studies that explore both technical and socio-technical challenges faced by developers (as also highlighted by De Stefano et al. (2022)). These hurdles include not only the technical demands of quantum software but also socio-technical issues such as the lack of collaboration and platform interoperability among quantum software engineers, further complicating the development and adoption of innovative technologies.

Always related to adoption challenges and socio-technical aspects, Zappin et al. (2025) recently published a paper investigating the challenges faced by developers into the development of *Hybrid Quantum-Classical applications*. Their findings revealed that developers face not only technical limitations related to hardware and libraries but, more critically, a range of socio-technical and cognitive challenges. These include frequent developer errors due to the steep quantum learning curve, difficulties in managing integration between classical and quantum components, and recurring problems induced by platform updates, undocumented behaviors, and insufficient support. The study introduces a comprehensive taxonomy of recurring issues and highlights the predominance of developer-driven faults, underscoring the urgent need for targeted debugging tools, clearer documentation, and educational resources tailored to the hybrid quantum-classical paradigm.

More directly related to our research, Stein et al. (2023) applied the principles of DoI theory (Rogers 2003) to various case studies involving the use of quantum computing, including applications in the pharmaceutical and finance sectors. Their findings confirmed that quantum computing remains an emerging and innovative technology, with DoI theory providing a useful framework for understanding its adoption. However, their study was limited to qualitative observations, as no statistical analysis was conducted, which restricts the generalizability of their findings and leaves room for further quantitative investigation.

In reviewing related works, numerous studies have highlighted the challenges that professionals encounter in adopting QSTs, urging the research community to address this gap with targeted research (De Stefano et al. 2022; Murillo et al. 2025; Shaydulin et al. 2020). While some preliminary efforts have been made, there remains a significant lack of rigorous, statistically grounded investigations into the factors that drive practitioners' intentions to adopt QSTs. Our article aims to fill this gap and make a valuable contribution to the field.

2.3 Technology Adoption Models

The adoption of technology has been extensively studied, leading to the formulation of various theories that identify variables (commonly referred to as factors or constructs) influ-

encing an individual's decision to adopt a technology. Among these theories, those of particular relevance to this study are the *Unified Theory of Acceptance and Use of Technology (UTAUT2)*, proposed by Venkatesh et al. (2012), and the *Diffusion of Innovations Theory (DoI)*, originally introduced by Everett Rogers in his seminal 1962 book "Diffusion of Innovations" (Rogers 2003). These theories have been extensively discussed in the context of software development and engineering, guiding the theoretical choices underpinning this research (Russo 2024; Pfleeger and Menezes 2000; Aversano et al. 2002; Voria et al. 2024; Lambiase and De Lucia 2024).

2.3.1 Extended Unified Theory of Acceptance and Use of Technology (UTAUT2)

Venkatesh et al. (2003) offered an in-depth analysis of eight key models and established the Unified Theory of Acceptance and Use of Technology (UTAUT). This framework suggests that four constructs are the main predictors of behavioral intentions (BI) and usage behavior (UB): the user's belief that the system will enhance job performance (Performance Expectancy, PE), the system's simplicity of use (Effort Expectancy, EE), the perception that necessary organizational and technical resources are available to support system usage (Facilitating Conditions, FC), and the influence of peers whose opinions matter (Social Influence, SI). Subsequently, Venkatesh et al. (2012) expanded the model to analyze technology acceptance and use in consumer settings by incorporating three additional constructs. The first, Hedonic Motivation, relates to how enjoyable or satisfying users find the system (HM). The second, Price Value, concerns the cognitive evaluation of the benefits versus the monetary costs of using the technology (PV). Lastly, Habit pertains to the extent to which individuals automatically engage in behaviors due to learning and routine (Hb). Further, it was hypothesized that individual factors such as age, gender, and experience might moderate the influence of these constructs on behavioral intentions and technology usage.

2.3.2 Diffusion of innovation Theory (DoI)

The theory of diffusion of innovation (DoI) provides insights by analyzing how innovations propagate successfully through a more refined measure of consumer behavior (Rogers 2003). According to DoI, factors influencing individual technology adoption include the ability of new technology to replace or complement existing ones, and its compatibility with the work environment and user needs. A key focus of this work is compatibility (Com), or the degree to which an innovation aligns with users' current values and needs. Subsequently to Rogers, personal innovativeness (Inn), described by Mun et al. (2006) as an "individual's willingness to try out a new IT," has been suggested as a significant variable affecting user technology acceptance.

2.3.3 Perceived Trust in Technology

Trust in technology is a critical factor in adoption, particularly in the context of information system technologies (Söllner et al. 2016; Müller et al. 2023). It can be defined as the potential feelings of uncertainty associated with using a technology. While not explicitly tied to any specific theoretical framework, trust is typically measured in terms of trust in the

technology itself (assessed through its reliability) and trust in the provider or entity supplying the technology (Söllner et al. 2016).

2.4 PLS-SEM in Software Engineering Studies

PLS-SEM is a statistical technique often compared to traditional regression methods, used to test and analyze complex relationships and to develop predictive models (Hair Junior et al. 2014). Unlike standard regression, PLS-SEM supports the analysis of intricate model structures, including multiple dependent variables and hierarchical constructs. It is particularly well-suited for exploratory research, especially in contexts with small sample sizes or when data violate assumptions required by other techniques. PLS-SEM is ideal for studying latent variables—constructs that cannot be directly observed and must be measured through validated instruments. Furthermore, it is widely used for theory testing, as it provides a broad range of statistical tools to assess not only the significance of hypotheses but also the quality, reliability, and validity of the measurement and structural models.

PLS-SEM is applied through a structured, theory-driven procedure. The process begins with the specification of both measurement and structural models, defining latent constructs and the hypothesized relationships among them based on prior theoretical frameworks. Measurement instruments are developed using validated scales from existing literature to ensure construct validity and reliability. Data are then collected from a target population, followed by a preliminary screening phase to handle missing values, assess response quality, and examine basic distributional properties. The measurement model is evaluated by analyzing internal consistency (e.g., Cronbach's alpha, composite reliability, ρ_A), convergent validity (via Average Variance Extracted), and discriminant validity (using the Heterotrait-Monotrait ratio). Once the measurement model satisfies quality criteria, the structural model is assessed by estimating path coefficients and testing their significance through bootstrapping techniques. Multicollinearity is checked using Variance Inflation Factors (VIF), while the model's explanatory power is evaluated via the coefficient of determination (R^2). Finally, direct, indirect, and total effects are examined to interpret the causal mechanisms linking constructs within the model.

In Software Engineering (SE), PLS-SEM is increasingly employed to analyze complex relationships involving latent constructs such as organizational culture, trust, or human factors. It is now recognized as a state-of-the-art technique for both theory development and predictive modeling in SE, although best practices and reporting standards are still maturing (Russo and Stol 2021). Its suitability for handling complex models, small samples, and non-normal data distributions makes it particularly valuable in empirical SE studies. Researchers use PLS-SEM to go beyond traditional metrics (e.g., lines of code), incorporating organizational and behavioral dimensions. Recent applications include studies on educational contexts (Che et al. 2021), software management and DevOps (Khattak et al. 2023), and the adoption of large language models (Russo 2024; Lambiase and De Lucia 2024; Choudhuri et al. 2024). Recent methodological work also emphasizes the need for standardized guidelines for specifying, estimating, and validating PLS-SEM models in SE.

3 Objective and Research Model

To address the previously described gaps, this research aimed to investigate the most influential factors in shaping software practitioners' adoption of QSTs. Consistent with established theories (Venkatesh et al. 2012, 2003), this study focuses on two interrelated dependent variables: the **intention to adopt QSTs (BI)** and the **frequency of their use (UB)**.

First, we developed a research model to map and represent the variables that, according to the theories outlined in Section 2.3, influence the intention to adopt and the actual use of QSTs. The model, presented in Fig. 1, integrates UTAUT2, the innovation characteristics of DoI, and perceived trust in technology to provide a more comprehensive understanding of QST adoption. This theoretical integration is based on three key assumptions: (1) QSTs are still in an emerging phase, with a consumer-driven adoption model, making their adoption a complex phenomenon that requires more than a single explanatory framework; (2) an integrative approach offers a more complete perspective on the causal mechanisms influencing adoption while generating insights that a single-theory model would not capture (Jackson et al. 2013); and (3) combining theories enhances the explanatory power and predictive strength of the results, reinforcing their validity.

Table 1 summarizes the hypotheses that are presented in the reminder of this section.

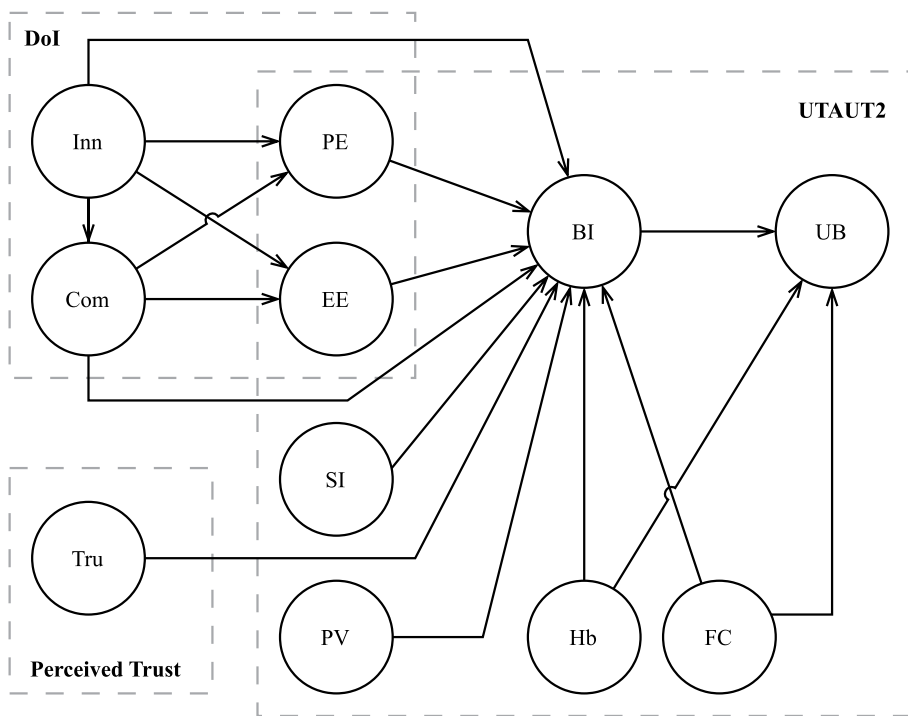


Fig. 1 Research model and hypotheses

Table 1 List of the hypotheses used to build the research model

ID	Relation	Description
H1	PE → BI	Performance Expectancy positively influences the behavioral intention to adopt QSTs
H2	EE → BI	Effort Expectancy positively influences the behavioral intention to adopt QSTs
H3	SI → BI	Social Influence positively influences the behavioral intention to adopt QSTs
H4	PV → BI	Price Value positively influences the behavioral intention to adopt QSTs
H5a	Hb → BI	Habit positively influences the behavioral intention to adopt QSTs
H5b	Hb → UB	Habit positively influences the actual use of QSTs
H6a	FC → BI	Facilitating Conditions positively influence the behavioral intention to adopt QSTs
H6b	FC → UB	Facilitating Conditions positively influence the actual use of QSTs
H7	BI → UB	Behavioral Intention positively influences the actual use of QSTs
H8a	Inn → Com	Innovativeness positively influences Compatibility
H8b	Inn → PE	Innovativeness positively influences Performance Expectancy
H8c	Inn → EE	Innovativeness positively influences Effort Expectancy
H8d	Inn → BI	Innovativeness positively influences the behavioral intention to adopt QSTs
H9a	Com → PE	Compatibility positively influences Performance Expectancy
H9b	Com → EE	Compatibility positively influences Effort Expectancy
H9c	Com → BI	Compatibility positively influences the behavioral intention to adopt QSTs
H10	Tru → BI	Perceived Trust in QSTs positively influences the behavioral intention to adopt them

3.1 UTAUT2 Constructs and Hypotheses

From the UTAUT2 model, six constructs were included in this study: performance expectancy (PE), effort expectancy (EE), social influence (SI), facilitating conditions (FC), habit (Hb), and price value (PV). The hedonic motivation construct was excluded, as it has been shown to be less effective in the adoption of IT technologies in work-related contexts. All the included constructs have been shown to be positively associated with an increase in the intention to adopt a technology, as they reflect how favorably the technology is perceived by individuals, albeit from different perspectives (Venkatesh et al. 2012; Russo 2024). Intention, in turn, is naturally linked to an increase in the actual use of the technology, serving as a mediator through which all other model variables exert their influence on this central construct (Venkatesh et al. 2012; Lambiase and De Lucia 2024). Additionally, habit and facilitating conditions have been found to directly impact actual technology use—bypassing behavioral intention (Venkatesh et al. 2012; Lambiase and De Lucia 2024).

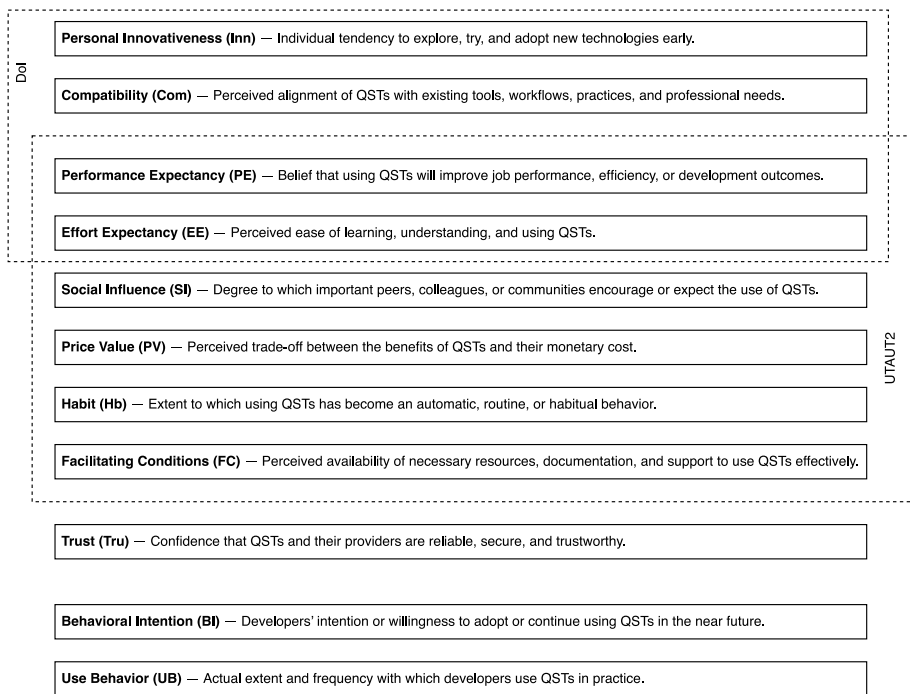


Fig. 2 Constructs and their definition

- H1 Performance expectancy positively influences the behavioral intention to adopt QSTs.
- H2 Effort expectancy positively influences the behavioral intention to adopt QSTs.
- H3 Social influence positively influences the behavioral intention to adopt QSTs.
- H4 Price value positively influences the behavioral intention to adopt QSTs.
- H5 Habit positively influences (a) the behavioral intention to adopt QSTs and (b) their actual use.
- H6 Facilitating conditions positively influences (a) the behavioral intention to adopt QSTs and (b) their actual use.
- H7 Behavioral intention to adopt QSTs positively influences the actual use of such technologies.

3.2 DoI Constructs and Hypotheses

Considering QSTs as a disruptive innovation, innovation-related factors play a significant role in shaping behavioral intention toward adoption. Given the overlap between some constructs in the DoI theory and those in UTAUT2, and to maintain model parsimony while avoiding unnecessary complexity in data collection, the focus was placed on two constructs less directly captured by UTAUT2: personal innovativeness (Inn) and compatibility (Com).

Innovativeness has been shown to be not only a significant direct predictor of the behavioral intention to adopt a new technology (Mun et al. 2006), but also an antecedent of compatibility, performance expectation, and effort expectation. The higher the level of

innovation of a user, the greater the propensity to feel compatible with the technology and to recognize the benefits of the technology.

H8 Developers with higher innovativeness levels have higher (a) compatibility, (b) performance expectancy, (c) effort expectancy, and (d) intention to adopt QSTs.

Compatibility has been shown to be a direct predictor of the intention to adopt a new technology and as an antecedent to performance expectancy and effort expectancy (Mun et al. 2006). It is particularly critical in SE, where developers must integrate new tools within existing development environments, workflows, and methodologies. In the case of QSTs, compatibility influences whether these technologies can be effectively used alongside classical software development frameworks.

H9 Developers with higher compatibility levels have higher (a) performance expectancy, (b) effort expectancy, and (c) the intention to adopt QSTs.

3.3 Perceived Trust Constructs and Hypotheses

Finally, since the current QST usage model, which is mainly based on the consumption of services provided by external entities, presents a high entry barrier, the research model also incorporates perceived trust (Tru) as a key construct. This construct captures multiple dimensions of trust: trust in the technology itself (in terms of reliability) and trust in the organizations providing these technologies. Therefore, perceived trust has a positive influence on the developers' intention to adopt QSTs.

H10 Perceived trust in QSTs positively influences the behavioral intention to adopt them.

4 Research Method and Hypothesis Testing

To test the hypotheses introduced in Section 3 and explore software practitioners' intentions to adopt and actual adoption of QSTs, we conducted a survey study combined with *Partial Least Squares Structural Equation Modelling (PLS-SEM)* (Hair Junior et al. 2014) for data analysis and statistical significance.¹

The research design involved the following steps:

1. We developed a questionnaire using validated scales from the literature to measure the constructs included in the research model described in Section 3.
2. We identified a set of filters that describe the ideal population for our work, i.e. software practitioners using QSTs.
3. We conducted a rigorous pre-screening and selection process to collect our participants sample.
4. We administered the questionnaire to software practitioners with experience in using QSTs to collect responses for measuring the constructs.
5. We analyzed the obtained data using PLS-SEM to test the hypotheses and evaluate the significance of each construct in predicting the intention to adopt and the actual usage of QSTs.

¹ We briefly introduced PLS-SEM in Section 2.4. For a detailed analysis, refer to the book of Hair Junior et al. (2014).

In order to conduct the empirical research in a well-designed manner, we adhered to the SIGSOFT Empirical Standards for Questionnaire Surveys (Ralph et al. 2020); we illustrate below examples of how these standards were concretely applied in our study:

- *Essential attributes* were systematically addressed. The target population was clearly defined, and a purposeful sampling strategy was implemented via the recruitment platform, as described in Section 4.3. Measurement instruments were derived from validated scales in the literature (Section 4.2), and underwent iterative pilot testing to ensure clarity and usability. Data preprocessing included rigorous procedures to handle missing data, detect anomalous response patterns, and identify outliers, ensuring methodological transparency and reproducibility (Section 5.1). All study materials, including the questionnaires, were shared in an online appendix to support open science and replicability (Lambiase and De Lucia 2025).
- *Desirable attributes* were also incorporated to strengthen the study’s methodological soundness. A detailed demographic profile of the sample is provided (Section 4.3.2). Ethical considerations were carefully addressed, with approval obtained from the relevant ethics committee and informed consent collected from all participants. Questionnaire layout, completion time, and clarity were refined through pilot testing. To mitigate response bias, we randomized the order of questions and embedded attention-check items.
- *Extraordinary attributes* were included to further enhance methodological rigor. In particular, we conducted construct validity analyses to confirm both convergent and discriminant validity of the measurement model (Section 5.2). Finally, we employed advanced statistical methods (PLS-SEM) to ensure robust validation of our theoretical model (Hair Junior et al. 2014).

The remainder of this section elaborates on the methodological steps undertaken in detail, focusing more on the specific guidelines adopted to ensure soundness.

4.1 Questionnaires and Data Collection

Data collection was conducted in two phases using two distinct questionnaires. The first, referred to as the “Pre-Screening Questionnaire,” aimed to identify suitable participants for the study. The second, referred to as the “Main Questionnaire,” was designed to measure the constructs in the theoretical model within the pre-screened population.

Both questionnaires were designed in accordance with the guidelines proposed by Kitchenham and Pfleeger (2008), which are widely adopted in software engineering survey research, as well as those by Andrews et al. (2007), which focus specifically on best practices for administering online questionnaires. To align with standard ethical practices in empirical research, the surveys were conducted anonymously.

Following Kitchenham and Pfleeger’s recommendations (Kitchenham and Pfleeger 2008), we clearly defined the objective of the study and framed it as a cross-sectional survey. Whenever possible, we relied on previously validated instruments from the literature rather than developing new ones, thus ensuring the reliability and validity of the measurement tools. Their guidelines also informed the layout and structure of the questionnaire: formatting (e.g., bold and italic) was used to highlight key elements, and questions were

logically grouped into thematic sections. The introductory section provided all necessary information to support participants in understanding and completing the survey.

The identification of the target population was achieved through pre-screening and filtering criteria. In line with Andrews et al.'s recommendations (Andrews et al. 2007), the recruitment platform was configured to: (1) ensure cross-device compatibility, (2) prevent duplicate submissions, (3) provide responsive layout across browsers and devices, and (4) allow participants to submit feedback upon completion.

The pre-screening phase began in November 2024 and required one week to gather a sufficient number of responses. Subsequently, the main questionnaire was administered to the pre-screened participants, also completed within a one-week.

An iterative pilot strategy was employed to (1) test the quality and clarity of the surveys and (2) estimate completion time. Initially, three pilot tests were conducted with 10 researchers from our network, and adjustments were made based on their feedback. Subsequently, a pilot for each questionnaire was conducted on the data collection platform. The Pre-Screening Questionnaire was administered to 5 qualifying participants, followed by the Main Questionnaire. The final pilot resulted in no further changes, and responses from pilot participants were included in the final dataset.²

4.2 Measurement Instruments

The research model included nine independent variables (six from UTAUT2, four from DoI, and one representing Perceived Trust) and two dependent variables: intention to use (*Behavioral Intention*, BI) and actual use (*Use Behavior*, UB) of QSTs. All variables were treated as latent constructs, meaning they were not directly observable and required measurement through reflective indicators. These reflective indicators were Likert-scale questions (henceforth referred to as items) included in the study's questionnaires.

The nine independent variable constructs were measured using items adapted from well-established sources. For UTAUT2 constructs, items were adapted from Venkatesh et al. (2012) and Rogers (2003); for DoI constructs, items were derived from Rogers (2003) and Mun et al. (2006). The items for Perceived Trust were self-developed, drawing on scales proposed by Choi and Ji (2015) and Nordhoff et al. (2021).³ The measurement was done following a reflective approach, in line with the papers that we used to select the measurement scales. All items were presented on a 7-point Likert scale ranging from “*Strongly disagree*” to “*Strongly agree*”, consistent with the original studies. Used items had demonstrated excellent reliability in prior studies, making them an ideal choice for our research context. Self-developed items for Perceived Trust were validated during pilot testing to ensure accuracy and clarity.

To measure the adoption and intention to adopt QSTs (dependent variables), using the established scales of Venkatesh et al. (2003) and Rogers (2003), the 6-point frequency scale was used to assess UB, while the 7-point Likert scale was retained for BI. This approach

² Ethical approval for this study was obtained from the ethical board of the University of Salerno (Italy).

³ The process of constructing the items involved a series of workshops conducted by the first author with the research group he was working with at the time of writing. Specifically, the first author, together with three other SE researchers (two of whom were researchers in the field of fairness and trust), read the scales proposed by Choi and Ji (2015) and Nordhoff et al. (2021) and developed and discussed the items used in the final version of the study.

ensured a robust and reliable evaluation of participants' intention to adopt and actual usage of QSTs.

In addition to the above, various demographic measures were also collected, such as gender, age, years of experience, and role. In order to avoid making questionnaires too long, the information was extracted from the recruiting platform we used. These measures allowed the recruitment phase to be balanced to ensure a heterogeneous, representative sample that is robust to influences from control variables.

4.3 Participants Recruitment and Selection

This section provides details on the participant recruitment process and the characteristics of the final sample.

4.3.1 Participants Recruitment

The recruitment and selection process targeted software practitioners actively involved in quantum software projects. To facilitate survey administration and data collection, Prolific,⁴ a platform tailored for academic research with built-in participant filtering capabilities, was utilized. Prolific is increasingly used in software engineering studies and has been demonstrated being a high-quality recruitment platform when specific criteria—like those we adhered to as outlined—are met (Eyal et al. 2021; Douglass et al. 2023).

A cluster sampling strategy was implemented to ensure that participants met predefined criteria aligned with the study's objectives. The ideal participant profile included professionals employed in the Information Technology sector, either full-time or part-time, who possessed programming skills and prior experience with QSTs. Participants whose use of QSTs was obligatory were excluded to prevent bias in responses related to intention-to-use factors. The recruitment process followed the protocol outlined by Alami et al. (2024), incorporating the following choices:

- The recruitment was conducted iteratively through pre-screening phases to refine participant selection.
- The questionnaire included questions to verify that respondents were developers (Danilova et al. 2021) and had relevant experience with QSTs.
- Compensation was tiered, offering £9 (“good”) remuneration for pre-screening and £12 (“great”) remuneration for the main survey.⁵

Participant experience with QSTs was assessed through a frequency-of-use scale and a verification question that required respondents to describe a quantum circuit. All responses underwent manual review to ensure accuracy and consistency, and these strategies were applied at multiple stages throughout the iterative questionnaires.

To further ensure participant reliability and comprehension, additional filters were applied. Respondents were required to:

⁴Prolific (www.prolific.com) [October 2025]

⁵Prolific provides a four-tier compensation framework—low, fair, good, and great—based on the amount of money you are paying your participants.

- be fluent in English (the survey language); and
- maintain a 100% approval rate on Prolific, with at least 25 completed surveys, ensuring a proven reliability.⁶

Responses that failed attention checks or were completed in less than half the estimated time were excluded.⁷ We also screened the dataset for AI-generated or low-quality responses. First, we used SurveyMonkey's integrated quality-check tool, which is specifically designed to flag responses exhibiting automated generation patterns. Second, we performed an additional statistical inspection to identify atypical patterns in response times, completion behaviour, and textual consistency. No responses were flagged through either procedure, and no indication of AI-generated content was found.

The minimum sample size was determined by conducting an a priori power analysis using G*Power (Faul et al. 2009) as recommended by Hair Junior et al. (2014). With an effect size of 15%, a significance level of 5%, and a power of 95%, the smallest required sample size for ten predictors was calculated to be 166 participants.

4.3.2 Participants Characteristics

Based on the criteria mentioned above, Prolific identified a potential pool of 5000 participants. To account for potential attrition between the two questionnaires, we initially collected 400 responses for the pre-screening survey; for the main one, we gathered 219 responses. After applying filtering for attention checks and time to completion, we obtained a final reliable set of 187 answers; these values are in line with similar studies involving similar theoretical models and objectives (Voria et al. 2024; Lambiase and De Lucia 2024; Russo 2024).

The analysis of participant roles indicates that Software Developer positions comprise the largest share, with 40.65% of participants. Software Engineers account for 26.02%, highlighting a substantial representation of technical engineering expertise within the sample. More specialized roles, such as Software Architects (7.32%) and Tester/QA Engineers (2.44%), are less prevalent, while senior leadership roles (e.g., CIO/CEO/CTO) are even rarer at 4.88%. In terms of tenure, most participants report 4-6 years (33.61%) or 1-3 years (32.79%) of experience in their current role, with a smaller proportion having over 9 years (21.31%) or within the 7-9 year range (9.02%). However, experience within the broader software industry skews more heavily toward seasoned professionals: the majority of participants report over 9 years of industry experience (34.96%), followed by 4-6 years (31.71%) and 1-3 years (21.14%).

More specifically, regarding the adoption of QSTs, 35.83% (67) of participants reported using them every day, 38.50% (72) reported using them frequently during the week, and 25.67% (48) indicated occasional monthly use. This distribution reflects a broad range of usage intensity among practitioners and, when combined with the demographic character-

⁶ On Prolific, a participant's approval rate reflects the percentage of accepted submissions, serving as a key indicator of reliability.

⁷ We used the SurveyMonkey platform to design and administer the survey. This tool enabled us to define internal logic rules that automatically excluded participants who failed any of our attention or reliability checks. This approach was necessary to reduce the manual effort required for validation and to prevent mistakenly awarding compensation on Prolific to participants who did not meet the required standards.

istics reported above and the rigorous filtering criteria applied during the screening phase, provides a robust and heterogeneous sample of QST users. While our sample does not aim to represent the entire population of software developers, it is representative of the target population of practitioners with actual experience in QSTs, which is required by the theoretical constructs used in our study.

4.4 Data Analysis

The data analysis process began with a *preliminary examination* to ensure data quality. This step involved checking for missing values, identifying suspicious patterns, detecting outliers, and examining data distribution. Although PLS-SEM is less restrictive than traditional methods, these preliminary checks were essential for validating data reliability.

Once validated, the dataset was imported into SmartPLS (Ringle et al. 2024) for PLS-SEM analysis. The analysis was conducted in two main phases: *Measurement Model Evaluation* and *Structural Model Evaluation*.

Measurement Model Evaluation focused on assessing the reliability and validity of the constructs. This phase aimed to ensure that the items used to measure each construct consistently and accurately reflected the underlying theoretical concepts, i.e., that we measured what we intended to measure. This phase involved four sub-phases:

- *Indicator Reliability*: Ensured that individual items consistently represented their respective constructs.
- *Internal Consistency Reliability*: Assessed using composite reliability to confirm the consistency of items within each construct.
- *Convergent Validity*: Verified that the indicators of each construct shared substantial variance, as measured by Average Variance Extracted (AVE).
- *Discriminant Validity*: Ensured that each construct was conceptually and statistically distinct from the others. *Structural Model Evaluation* analyzed the relationships between constructs and tested the hypothesized paths. This phase evaluated the explanatory and predictive capabilities of the model. The following aspects were assessed:
- *Collinearity*: Checked for multicollinearity among predictors using Variance Inflation Factor (VIF).
- *Significance of Relationships*: Tested the significance of both direct and indirect hypotheses using bootstrapping.
- *Explanatory Power*: Measured the model's ability to explain variance in the dependent variables using R-squared values.
- *Predictive Power*: Assessed the model's ability to predict outcomes using measures such as Q-squared.

A key focus of this study was on the dependent variables—*intention to use* and *actual use* of QSTs—while ensuring all hypotheses were comprehensively tested. In addition to analyzing direct relationships, particular attention was given to the *total effect* of constructs on the dependent variables. This approach allowed for the examination of both direct and indirect effects. For instance, while PE might not have a direct link to UB, its effect was tested through the intermediate role of BI (PE → BI → UB).

5 Results

The following section presents the results obtained from the PLS-SEM analysis, offering insights into the structural model's path coefficients, the measurement model's reliability and validity, and the overall model fit.⁸

5.1 Preliminary Analysis

Before proceeding with the main PLS-SEM analysis, a preliminary data examination was conducted to ensure data quality and integrity. Instances of missing values were minimal, with only 20 cases observed, likely attributed to the high quality of the questionnaire design and the reliability of the sample, ensured by the Prolific approval rate filter. These missing values accounted for less than 0.5% of the total responses per item and did not pose any significant issues, as SmartPLS is equipped to handle missing data automatically. Additionally, the dataset was reviewed for suspicious response patterns, and none were identified.

Data distribution was also examined for skewness and kurtosis to assess normality. As a general guideline, values exceeding the range of -2 to +2 indicate nonnormality. The analysis revealed slight deviations from normality in three indicators: FC2 (2.49), Tru5 (2.18), and SI3 (2.09). However, since normality is not a requirement in PLS-SEM and the deviations were minor, the indicators were retained for subsequent analysis. This decision ensured that the dataset remained comprehensive while adhering to the methodological flexibility of PLS-SEM.

5.2 Measurement Model Evaluation

As a first step in the evaluation of the theoretical model, it is paramount to evaluate the reliability of the constructs of the model (Hair Junior et al. 2014).

5.2.1 Indicator Reliability

The first step in assessing the measurement model, as outlined by Hair Junior et al. (2014), is to evaluate the reliability of the indicators by examining their *outer loadings*. High outer loadings suggest that the indicators capture a substantial proportion of variance in the construct they represent. The generally accepted guideline is to retain indicators with outer loadings above 0.708, as these indicate that more than 50% of the variance is shared with the construct. Indicators with loadings below 0.40 are typically removed, while those falling between 0.40 and 0.70 are considered for removal if doing so improves internal consistency reliability or convergent validity.

The outer loadings for all indicators are available in the online appendix (Lambiasi and De Lucia 2025). Two indicators, SI4 and Inn3, had outer loadings below 0.70 (0.61 and 0.50, respectively). However, since no indicator exhibited a loading below 0.40 and retain-

⁸Following the suggestion of one of the reviewers, we have added Fig. 2, which shows the definition of the constructs in order to assist the reader in interpreting the results.

Table 2 Measurement Model Evaluation Metrics

Construct	α	ρ_A	ρ_c	AVE
Performance Expectancy (PE)	0.854	0.860	0.895	0.632
Effort Expectancy (EE)	0.910	0.910	0.937	0.788
Social Influence (SI)	0.814	0.844	0.870	0.577
Price Value (PV)	0.902	0.908	0.953	0.911
Habit (Hb)	0.905	0.916	0.940	0.840
Facilitating Conditions (FC)	0.847	0.859	0.897	0.685
Innovativeness (Inn)	0.641	0.756	0.786	0.563
Compatibility (Com)	0.796	0.806	0.880	0.711
Perceived Trust (Tru)	0.756	0.768	0.890	0.803
Behavioral Intention (BI)	0.847	0.847	0.907	0.765

ing these items did not negatively affect internal consistency or validity, all indicators were retained for further analysis.⁹

5.2.2 Internal Consistency Reliability

The second step involved assessing *internal consistency reliability* to ensure that the indicators consistently measure their respective constructs. This evaluation was conducted using three key metrics: *Cronbach's alpha* (α), *composite reliability* (ρ_c), and the *reliability coefficient* (ρ_A).

As shown in Table 2, all constructs exceeded the recommended threshold of 0.60 for these measures, as outlined by Hair Junior et al. (2014). These results confirm that the indicators demonstrate sufficient internal consistency, providing a robust foundation for further analysis.

5.2.3 Convergent Validity

Convergent validity measures the extent to which indicators of a construct correlate positively with one another, reflecting the shared variance among the indicators of the same construct (Hair Junior et al. 2014). Given that all constructs in our model are measured using reflective indicators, it was expected that the indicators would converge and exhibit substantial shared variance. The primary metric used to evaluate convergent validity is the *average variance extracted* (AVE), where an AVE value of 0.50 or higher indicates that the construct explains more than half of the variance of its associated indicators.

As shown in Table 2, all constructs in our model achieved AVE values above the 0.50 threshold, demonstrating strong convergent validity and supporting the robustness of the measurement model.

5.2.4 Discriminant Validity

The final test focused on assessing discriminant validity, which evaluates the extent to which a construct is conceptually and empirically distinct from others. Given that our model

⁹For Inn, the full model (with Inn3) shows $\alpha = 0.641$, $\rho_A = 0.756$, $\rho_c = 0.786$, AVE= 0.563, whereas removing Inn3 yields $\alpha = 0.649$, $\rho_A = 0.696$, $\rho_c = 0.847$, AVE= 0.736; since the removal does not improve reliability and substantially reduces AVE, we retained Inn3.

integrates constructs from two closely related theories (UTAUT2 and DoI), a certain degree of similarity among constructs was anticipated. This overlap stems from the use of items adapted from both the scales of Venkatesh et al. (2012) and Rogers (2003).

As suggested by Hair Junior et al. (2014), we used the *heterotrait-monotrait ratio* (HTMT) as a reliable criterion for assessing discriminant validity; a value above 0.90 suggests insufficient discriminant validity, values between 0.85 and 0.90 are considered acceptable, and values below 0.85 are deemed excellent. Most of the HTMT values were below the recommended threshold of 0.85, with only three slightly exceeding this limit, i.e., FC and Com (0.868), PE and BI (0.891), and PE and Com (0.870). We also verified the bootstrap intervals of the HTMT to verify if one falls between the two confidence intervals, as suggested by Henseler et al. (2015); no pair was characterized by this, reinforcing the reliability and validity of the measurement model.

5.3 Structural Model Evaluation

Following the evaluation of the measurement model, the structural model was assessed to examine the relationships between constructs and validate the hypotheses.

5.3.1 Collinearity Analysis

The first step in structural model evaluation involves analyzing collinearity between exogenous and endogenous variables to ensure accurate path estimations. Collinearity, if present, can distort regression coefficients and weaken the validity of the results. To detect multicollinearity, the *Variance Inflation Factor* (VIF) was employed—a widely recognized metric in regression analysis. A VIF value below 3 is generally preferred, while values under 5 are considered acceptable in most cases.

In our analysis, the majority of VIF values were below the ideal threshold of 3, ensuring minimal collinearity among variables. However, four paths (i.e., Hb, Com, BI, and PE with BI) had VIF values slightly exceeding 3 but remained well below the acceptable upper limit of 5. These results indicate that multicollinearity is not a significant concern in the structural model.

5.3.2 Significance and Relevance of the Hypotheses

The structural model analysis assessed the significance and relevance of the hypothesized relationships. Significance testing was conducted using the bootstrapping method with 10 000 sub-samples, following the recommendations of Hair Junior et al. (2014). The analysis provided *T-statistics*, *p-value*, and significance levels for each hypothesized relationship.

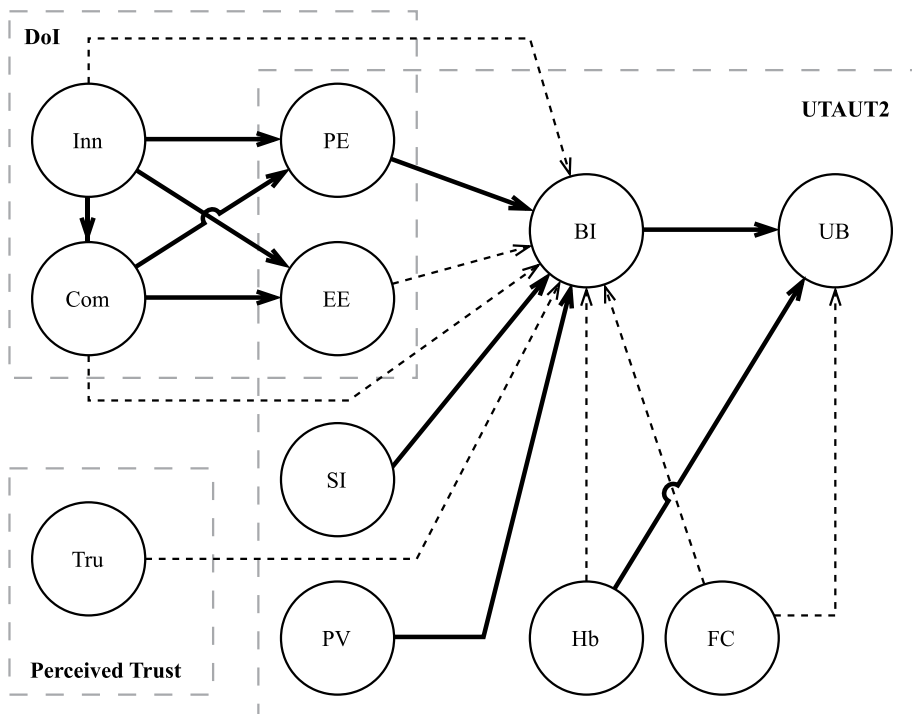
The results indicate several significant relationships within the model, with 9 out of 17 hypotheses found to be supported, as shown in Table 3 and Fig. 3. Notably, BI exhibited a significant relationship with UB, while Hb showed significant relationships with both BI and UB. PE demonstrated a significant relationship with BI, emphasizing its role within the structural model. Additionally, SI and PV were identified as having significant relationships with BI, contributing to the understanding of factors influencing intentions.

To evaluate the relevance of the significant relationships, standardized path coefficients were analyzed. PE was identified as having one of the strongest relationships with BI, fol-

Table 3 Significance and Rel-
evance of the Hypotheses

ID	Hypotheses	Path Coefficient	T statistic	p-value	Sig- nifi- cance
H1	PE → BI	0.395	4.778	0.000	***
H2	EE → BI	0.169	1.907	0.057	.
H3	SI → BI	0.148	1.992	0.046	*
H4	PV → BI	-0.195	2.842	0.004	**
H5a	Hb → BI	-0.006	0.072	0.943	
H5b	Hb → UB	0.402	5.271	0.000	***
H6a	FC → BI	0.111	1.329	0.184	
H6b	FC → UB	0.137	1.714	0.087	.
H7	BI → UB	0.197	2.744	0.006	**
H8a	Inn → Com	0.350	5.670	0.000	***
H8b	Inn → PE	0.280	5.513	0.000	***
H8c	Inn → EE	0.289	4.436	0.000	***
H8d	Inn → BI	0.066	1.176	0.240	
H9a	Com → PE	0.625	14.069	0.000	***
H9b	Com → EE	0.546	8.043	0.000	***
H9c	Com → BI	0.149	1.509	0.131	
H10	Tru → BI	0.101	1.250	0.211	

Note: .: $p < 0.1$; *,
 $p < 0.05$; **, $p < 0.01$; ***,
 $p < 0.001$

**Fig. 3** Significant hypotheses (highlighted in bold)

lowed closely by Hb, which also demonstrated a strong relationship with UB. Furthermore, the results highlighted that BI mediated the relationships between several constructs and UB, reinforcing its central role in the structural model. Constructs such as SI and PV exhibited moderate yet significant relationships with BI.

5.3.3 Total Effect Analysis

The analysis of the total effects highlights the relationships between the constructs and the dependent variables, BI and UB. Table 4 presents the total effects, *p-values*, and significance levels; Fig. 4 represent the same information in a graphical way. These total effects encompass both direct and indirect relationships, providing a comprehensive view of the overall impact of each construct.

Table 4 Significance and Relevance of the Total Effect (Tot. Eff.)

Construct	BI		UB	
	Tot. Eff.	p-value	Tot. Eff.	p-value
Behavioral Intention (BI)			0.197	0.006 (**)
Compatibility (Com)	0.488	0.000 (***)	0.096	0.016 (*)
Effort Expectancy (EE)	0.169	0.057 (.)	0.033	0.127
Facilitating Conditions (FC)	0.111	0.184	0.158	0.039 (*)
Habit (Hb)	-0.006	0.943	0.400	0.000 (***)
Innovativeness (Inn)	0.396	0.000 (***)	0.078	0.020 (*)
Performance Expectancy (PE)	0.395	0.000 (***)	0.078	0.023 (*)
Price Value (PV)	-0.195	0.004 (**)	-0.038	0.033 (*)
Social Influence (SI)	0.148	0.046 (*)	0.029	0.131
Trust (Tru)	0.101	0.211	0.020	0.252

Note: $\cdot$: $p < 0.1$; *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$

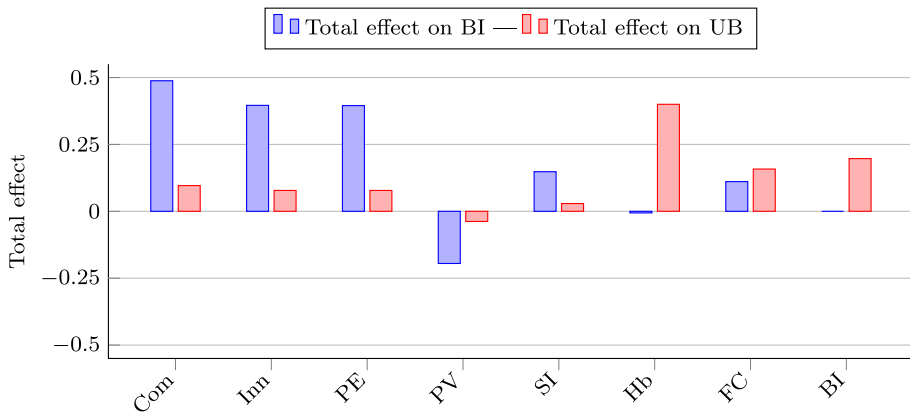


Fig. 4 Summary of total effects of the main constructs on adoption intention (BI) and actual use (UB) of QSTs. Positive bars indicate a positive overall influence, negative bars indicate a negative overall influence. Significance levels follow Table 4

Regarding the intention to adopt QSTs (BI), Com exhibited the strongest total effect, confirming its central role in shaping intentions. Inn and PE also demonstrated significant and substantial total effects, emphasizing their importance as key drivers of intention within the model. SI showed a significant total effect, indicating that social influence meaningfully influences intention. EE and Tru showed weaker, non-significant effects, suggesting limited contributions. Hb and PV demonstrated negligible or negative effects.

Regarding the frequency of use (UB), Hb emerged as the most influential predictor, underscoring the dominant role of habitual behavior. BI also had a significant effect, reinforcing its mediating role. Com and FC showed moderate effects, reflecting their indirect influence on behavior. PV showed a small but significant negative effect, indicating an interesting result. Constructs such as EE, SI, and Tru had non-significant effects, suggesting limited influence in this context.

These findings highlight Hb and BI as primary determinants of the actual use of QSTs (UB) while illustrating the varied roles of other constructs in shaping behavioral outcomes.

5.3.4 Explanatory Power

The final phase of the analysis evaluated the model's explanatory capability, focusing on its ability to fit the data and quantify the strength of relationships within the model. The coefficient of determination (R^2), which measures the proportion of variance explained by the model, was used for this assessment. R^2 values range from 0 to 1, with higher values (greater than 0.19) indicating greater explanatory power (Raithel et al. 2012; Hair Junior et al. 2014).

The analysis revealed an R^2 value of 0.669 for BI and 0.431 for UB, indicating that the model explains 66% of the variance in the intention to use QSTs and 43% of the variance in their actual usage. Additionally, the analysis of Effort Expectancy and Performance Expectancy, both predicted by Innovativeness and Compatibility Factors, showed R^2 values of 0.492 and 0.592, respectively. These results highlight the model's robust explanatory power across key constructs.

5.3.5 Predictive Power

As previously noted, this study represents the first attempt to investigate the adoption of QSTs by integrating multiple theoretical frameworks. This ensemble approach was chosen to capture a broader range of influencing factors, laying the groundwork for future research and providing foundational insights into the phenomenon. It is worth noting that, when combining constructs from different theories, predictive power—typically used to assess managerial applicability—often plays a secondary role. Prior studies have shown that single-theory models generally offer higher predictive performance than theoretical ensembles. Nonetheless, in the interest of methodological completeness, we report below the results of the predictive power analysis to offer the reader a comprehensive view of the proposed model.

To assess the model's utility for managerial decision-making, we tested whether the results of our PLS-SEM algorithm are applicable beyond the data used in the estimation process. We employed the PLS_{predict} procedure (Shmueli et al. 2016), which involves separating the overall dataset into training and holdout samples. The measures used in this analy-

sis include Stone-Geisser's Q^2 statistic, the *mean absolute error* (MAE), and the *root mean square error* (RMSE). These statistics were tested against a benchmark, with Shmueli et al. (2016, 2019) proposing the use of linear regression model (LM) results as the benchmark.

Positive Q^2 values indicate that the model's prediction error is smaller than that of the benchmark. Additionally, lower values of MAE and RMSE suggest that the model possesses high predictive power. Table 5 reports the values for the dependent variables in the model. Q^2 values were positive for all the dependent variable items, indicating meaningful predictive relevance. Moreover, all variables exhibit extremely similar values compared to the benchmark, indicating that the PLS-SEM model demonstrates high predictive power and thus supporting its utility for practical, data-driven decision-making (Table 6).

Summary of the Results.

The analysis confirms that the model is reliable and well-suited to predict which factors drive interest in adopting QSTs. The results show that Hb and BI are the strongest predictors of UB, highlighting the role of routine and intent in adoption. PE, Com, and Inn significantly influence BI, while PV negatively impacts adoption. Surprisingly, EE and Tru had no significant effect. SI moderately influenced BI but did not translate into UB.

6 Discussion

In this section, we discuss our findings in depth. We begin by highlighting the main contributions of our work in relation to the existing literature presented in Section 2 (Section 6.1). Next, we examine each hypothesis introduced in the results section and offer potential interpretations of the observed outcomes (Section 6.2). Then, we outline a set of recommendations for practitioners involved in the development and adoption of QSTs, and propose future research directions based on our findings (Section 6.3). Finally, we report on the threat to validity of the study (Section 6.4).

6.1 Advancements Concerning Related Works

As discussed in Section 2, our work represents one of the first empirical study aimed at investigating the factors that drive software engineers to adopt QSTs. This focus on socio-technical aspects within the developer community is not incidental, but rather responds to the needs highlighted by De Stefano et al. (2022) and Murillo et al. (2025). Accordingly, our

Table 5 Prediction Summary (PLS) and Comparison with Linear Regression (LM)

Item	Q^2	PLS_RMSE	PLS_MAE	LM_RMSE	LM_MAE
BI1	0.278	0.789	0.617	0.717	0.559
BI2	0.192	1.085	0.822	0.973	0.727
BI3	0.251	0.810	0.617	0.712	0.558
UB1	0.385	0.823	0.672	0.860	0.691

Table 6 Summary of hypotheses, significance, and practical implications for practitioners

Hypothesis	Supported	Practical Implication
H1: PE $\rightarrow$ BI	Yes	Emphasize concrete productivity gains when presenting QSTs to developers; performance benefits strongly drive intention.
H2: EE $\rightarrow$ BI	No	Simplifying tools alone does not raise adoption intention; abstraction layers must be paired with clear value propositions.
H3: SI $\rightarrow$ BI	Yes	Community advocacy and peer endorsement help generate interest; social diffusion mechanisms matter in early adoption.
H4: PV $\rightarrow$ BI	Yes (negative)	Lower-cost or free tools are not necessarily preferred; developers associate higher price with higher reliability and capability.
H5a: Hb $\rightarrow$ BI	No	Habit does not increase intention; exposure alone is insufficient without perceived benefits.
H5b: Hb $\rightarrow$ UB	Yes	Routine usage is the strongest predictor of actual adoption; organizations should foster repeated exposure.
H6a: FC $\rightarrow$ BI	No	Supportive environments do not raise intention directly; training and resources matter more at usage stage.
H6b: FC $\rightarrow$ UB	Yes (weak)	Better documentation, tutorials, and tool support increase actual usage in practice.
H7: BI $\rightarrow$ UB	Yes	Intention translates into behavior; targeting motivation improves real adoption.
H8a: Inn $\rightarrow$ Com	Yes	Innovative developers more easily see QSTs as compatible; early adopters remain a key user base.
H8b: Inn $\rightarrow$ PE	Yes	Innovative users expect larger performance gains; target this group with advanced use cases.
H8c: Inn $\rightarrow$ EE	Yes	Innovative users perceive QSTs as less difficult; outreach to “innovation-prone” developers is effective.
H8d: Inn $\rightarrow$ BI	No	Innovativeness alone doesn’t create intention unless value and compatibility are clear.
H9a: Com $\rightarrow$ PE	Yes	Improving integration with existing workflows boosts perceived performance benefits.
H9b: Com $\rightarrow$ EE	Yes	Tools aligned with developers’ environments feel easier to use; interoperability is essential.
H9c: Com $\rightarrow$ BI	No	Compatibility influences intention only indirectly (through PE); highlight real performance impact.
H10: Tru $\rightarrow$ BI	No	Trust in providers or platforms is not decisive for early adopters; focus on capability, not reliability messaging.

study constitutes a clear advancement over several other existing related works on socio-technical aspects.

Our study echoes the concerns raised by Bisicchia et al. (2024) regarding the technical complexity and fragmentation of the quantum software landscape. They identify the lack of high-level abstractions and platform interoperability as key obstacles to widespread adoption. Our empirical results confirm these challenges from the perspective of practitioners: constructs such as compatibility, facilitating conditions, and trust significantly influence adoption intention and use. However, we go further by providing a validated theoretical model that quantifies the impact of these factors through structural equation modeling. While Bisicchia et al. (2024) propose future research directions such as virtual quantum providers and multi-stack pipelines, our contribution lies in demonstrating that current adoption barriers are already measurable and influential in real-world usage. In doing so, we shift the discussion from conceptual diagnosis to empirical explanation, grounding their propositions in behavioral evidence from the developer community.

Our work complements and extends the study by Zappin et al. (2025), who qualitatively analyzed over 500 developer forum threads to identify recurring challenges in hybrid quantum-classical application development. Their taxonomy highlights common issues such as platform incompatibilities, insufficient documentation, and a high incidence of developer mistakes—particularly in configuring environments, using quantum libraries, or implementing embedding strategies. Our findings empirically confirm several of these points: for instance, we show that facilitating conditions (e.g., availability of training resources and toolchains) and habit (routine exposure to QSTs) are statistically significant predictors of actual usage, supporting their claim that better developer support reduces friction. However, our contribution advances the state of the art in three key ways. First, while Zappin’s analysis remains descriptive, we introduce and validate a theory-driven model of QST adoption, integrating UTAUT2, DoI, and perceived trust, and test it quantitatively on 187 QST practitioners. Second, we move beyond issue classification by identifying causal pathways: for example, we show that compatibility and innovativeness increase performance expectancy, which in turn drives intention to adopt. Third, we uncover unexpected dynamics not captured in prior work—most notably, a negative total effect of price value, suggesting that open access and low-cost entry points may not correlate with long-term engagement. In sum, while Zappin et al. (2025) offer a valuable empirical map of developer struggles, our study transforms these insights into a validated model that explains why adoption occurs, and under which conditions.

Our study complements and extends the work by Stein et al. (2023), who applied DoI theory to analyze early corporate adoption of quantum computing through qualitative case studies. While their work identifies innovation dynamics across four industries, our contribution is twofold. First, we move from descriptive analysis to causal inference by quantitatively modeling key adoption drivers—such as performance expectancy, compatibility, and personal innovativeness—through a validated PLS-SEM model. Second, we focus on the practitioner level rather than organizational strategy, revealing that intention does not automatically lead to usage: only habit and facilitating conditions significantly predict actual adoption. Moreover, we identify a counterintuitive negative effect of price value on intention, an aspect not captured in prior work. By integrating UTAUT2, DoI, and trust, our model provides a theory-grounded and generalizable explanation of QST adoption, complementing existing macro-level insights with a nuanced, socio-technical perspective on developer behavior.

6.2 Discussion on Hypotheses Significance and Interpretation of Findings

In this section, we discuss the main results of our investigation focusing on their interpretations (Table 6).

6.2.1 On the Exploratory and Predictive Power of the Proposed Research Model

An important aspect to highlight is that the theoretical model developed—rooted in Jackson’s perspective that innovative technologies should be investigated using complex and broader models (Jackson et al. 2013)—demonstrates excellent explanatory power for the two primary variables of interest, intention to adopt (BI) and frequency of adoption (UB), as well as for other dependent variables, such as the expected effort (EE) and expected performance in a

working environment (PE). This outcome not only supports Jackson's proposition but also reinforces the effectiveness of UTAUT2 and DoI in explaining technology adoption, even in the context of software development.

Beyond explanatory power, the analysis of the model's predictive power also yielded promising results. This suggests that our model is not only effective in describing relationships within the observed dataset, but also generalizes well to unseen data. Consequently, it can serve as a valuable tool for informing managerial decisions related to the adoption of QSTs. In the remainder of this section, we translate these findings into actionable recommendations for practitioners.

6.2.2 Compatibility, Personal Innovativeness, and Performance Expectancy as Key Predictors

The total effect analysis identified the compatibility of the technology with personal needs (Com) and the tendency to explore innovative technologies (Inn) as the primary predictors of the intention to adopt QSTs (BI). This result leads to two distinct interpretations. First, the current adopters of QSTs are predominantly individuals with a high inclination toward innovation—an indication that the technology is still widely perceived as emerging. Second, the relevance of compatibility suggests a shift in how QSTs are being approached: beyond novelty, developers are beginning to evaluate these technologies based on how well they fit into real-world contexts and address concrete professional needs.

However, this seemingly straightforward result reveals a deeper insight. While the total effects of compatibility and innovativeness are significant, their direct influence on behavioral intention is not. Instead, their impact is fully mediated by performance expectancy (PE)—that is, by developers' expectations regarding how QSTs can improve their job performance. This reinforces the idea that perceived utility, especially when aligned with developers' actual workflows and goals, has become central to adoption decisions.

A further consideration grounded in the nature of QSTs helps contextualize these findings. Because quantum technologies are still in a rapidly evolving phase, their innovative character itself represents a perceived professional advantage: early familiarity with QSTs is often seen by developers as strategically valuable for future career opportunities, especially given the growing interest in areas such as quantum machine learning and quantum-enhanced AI. At the same time, the increasingly frequent appearance of plausible quantum-related use cases—particularly for problems involving complex optimization or high-dimensional data—may amplify developers' expectations that QSTs will soon offer tangible performance advantages in practice. Similarly, the role of compatibility can also be interpreted through the lens of real problem-solving needs. Early adopters may perceive QSTs as promising tools for tackling computationally intensive problems more efficiently, even if current NISQ-era hardware imposes concrete limitations. This forward-looking expectation likely strengthens the perceived alignment between QST capabilities and developers' long-term professional or organizational goals, thereby contributing to the observed effects in our model.

6.2.3 Social Influence and Intention to Adopt

Another noteworthy finding, albeit statistically weaker, is the relationship between peer opinion (SI) and intention to adopt QSTs (BI). The opinions of those surrounding develop-

ers—their professional and community networks—positively correlate with their intention to adopt QSTs. However, this influence does not translate into actual adoption, as social influence does not significantly affect the frequency of use when mediated by the intention to use. This suggests that when transitioning from intention to actual adoption, other variables play a more decisive role.

This result becomes clearer when considering the specific dynamics of the QST domain. Quantum computing is strongly shaped by open-source communities and advocacy groups that actively promote experimentation, learning, and early engagement with quantum technologies (Shaydulin et al. 2020; Unitary Fund 2024). These communities often amplify the perceived importance of “being part of the quantum wave,” motivating practitioners to express interest or intention to adopt QSTs. Yet, when developers attempt to translate this social encouragement into concrete use, they face domain-specific barriers such as restricted access to quantum hardware, the maturity limits of NISQ-era devices, and the scarcity of stable, production-ready use cases. These constraints likely weaken the progression from social influence to actual behavior, explaining why SI affects intention but not usage in our model.

6.2.4 Facilitating Conditions and Habit as Drivers of Adoption

A contrasting result to SI emerges with Facilitating Conditions (FC) and Habit (Hb), which significantly predict UB but do not influence BI. A supportive environment—one that provides technological and educational resources—directly influences usage, suggesting that developers may adopt QSTs even without strong personal interest. This finding aligns with the current complexity of QSTs, which still operate at a low level of abstraction, often requiring developers to engage directly with quantum circuits and related concepts—elements that remain closely tied to quantum physics. As such, further effort is needed to raise the abstraction level of QSTs.

This interpretation becomes even clearer when considering the current maturity of the QST ecosystem. NISQ-era devices offer limited qubit counts, noisy executions and constrained access to hardware, while many tools still lack stable APIs, comprehensive documentation, or higher-level programming abstractions. These limitations increase the importance of facilitating conditions: developers rely heavily on tutorials, cloud credits, simulators, community support, and platform documentation to be able to use QSTs in practice. As quantum tooling gradually evolves—through improved SDKs, middleware, hybrid workflows and more accessible educational resources—the impact of facilitating conditions may become even more pronounced, enabling broader adoption beyond early adopters. Our results therefore highlight not only the current reliance on support structures, but also the critical role that ecosystem growth will play in shaping future usage patterns.

6.2.5 The Counterintuitive Role of Price Value

An intriguing finding concerns Price Value (PV)—the trade-off between the perceived benefits of a technology and its financial cost—and its relationship with our two adoption proxies (BI and UB). Typically, higher PV is associated with increased adoption (Venkatesh et al. 2012), yet in this study we observe the opposite.

While counterintuitive, this result likely reflects the current landscape of QST offerings. On one hand, expensive or paid quantum technologies are often perceived as more robust, better maintained, or more “enterprise-ready,” which may signal greater value and maturity to practitioners (Juhana 2023; Hariss et al. 2020). This perception may outweigh cost considerations in an emerging domain where reliability is scarce.

On the other hand, low-cost or free alternatives—typically open-source—tend to impose practical limitations. Many of these tools restrict developers to simulator-only environments or provide access to quantum hardware with very limited qubit counts, long queue times, or constrained execution capacity (Floyd et al. 2007; Rathee and Chobe 2022). Free platforms often offer fewer abstractions, limited documentation, and reduced developer support, increasing the effort required to use them and lowering perceived value despite the absence of cost. These constraints reduce the attractiveness of no-cost solutions for practitioners who wish to explore realistic or production-oriented quantum scenarios.

Moreover, the cloud-based nature of nearly all QST platforms introduces additional costs or resource tiers. Although some solutions, such as Qiskit (Javadi-Abhari et al. 2024), offer discounted access for researchers, individual developers outside academic or industrial programs may face substantial costs to run meaningful experiments on actual quantum hardware. In this sense, higher cost may correlate with broader access, better tooling, and more stable execution environments.

6.2.6 On the Marginal Role of Effort Expectancy and Trust

A surprising result of our analysis is the non-significant role of effort expectancy (EE) and perceived trust (Tru) in predicting the intention to adopt QSTs. While both constructs are frequently significant in technology adoption studies, their lack of influence in this context suggests that QST adoption may follow different mechanisms during early-stage diffusion (Rogers 2003).

Regarding the nonsignificant role of trust, early adopters in highly innovative domains tend to acknowledge that emerging technologies are inherently evolving, making trust a secondary concern (Rogers 2003). Rather than focusing on reliability, they prioritize potential benefits and long-term improvements of the technology. This is consistent with practitioner reports showing that QST ecosystems remain immature, rapidly evolving, and affected by tooling instability (De Stefano et al. 2024b; Murillo et al. 2025). At this stage, confidence in the technology itself is therefore less relevant than the perceived value it may offer in practical applications.

Similarly, effort expectancy does not significantly influence adoption decisions. Developers in technical fields are accustomed to working with highly complex tools, and community-driven resources often mitigate concerns about the learning curve. Moreover, early adopters frequently adopt new technologies incrementally, experimenting through small-scale implementations before integrating them more systematically. This reduces cognitive barriers despite the documented complexity of QSTs (Bisicchia et al. 2024).

Importantly, these findings do not imply that QSTs are easy to adopt—on the contrary, their complexity is widely reported in the literature (Murillo et al. 2025; Zappin et al. 2025). However, the key insight is that perceived difficulty does not currently serve as a decisive factor in adoption choices. One plausible explanation is that most available QST tools operate at uniformly low levels of abstraction, such as direct manipulation of circuits or hybrid

classical–quantum workflows (Ali 2024; Long and Zhao 2024). When all tools share comparable effort requirements, differences in perceived effort become less meaningful predictors.

Additionally, the still-limited number of QST platforms constrains developers to whichever solution offers the required functionality, regardless of ease of use. This lack of choice—also highlighted in prior studies on current QST ecosystems (Bisicchia et al. 2024; Zappin et al. 2025)—reduces the explanatory power of EE in this phase of technology diffusion.

6.3 Recommendations for Practitioners and Future Research Directions

In the following, we present recommendations for practitioners and future research directions based on the analysis of the hypotheses (Section 6.2) and the limitations of the study (Section 6.4).

6.3.1 Recommendations for Practitioners

The results of our study suggest several concrete recommendations for professionals involved in the design, evaluation, and adoption of QSTs.

👉 **Adopt validated models to assess adoption readiness and design onboarding processes.** Practitioners can use our theoretical frameworks to evaluate the openness of developers to quantum technologies (as demonstrated by the results of the structural model evaluation phase). These insights can inform hiring strategies and the design of onboarding programs tailored to reduce resistance and accelerate adoption.

👉 **Specialize QST platforms for defined use cases through concrete tool support.** Our findings confirm the importance of compatibility in shaping adoption decisions. Tool providers should therefore prioritize domain-specific solutions aligned with concrete use cases – not only in abstract terms, but concretely providing high-level abstractions, templates and workflow support. For example, the Qiskit Optimization module¹⁰ enables developers to express combinatorial or quadratic optimization problems through a high-level API (e.g., ‘QuadraticProgram’), automatically handling conversion to QUBO/Ising formulations and enabling execution via quantum or hybrid solvers, thus sparing developers from low-level circuit programming. Similarly, libraries like PennyLane¹¹ support variational quantum algorithms (e.g., VQE, QAOA) and quantum-classical machine learning pipelines with abstractions compatible with classical ML frameworks, lowering the barrier for data scientists and ML practitioners. By offering ready-to-use, domain-oriented tools with stable APIs and clear, reproducible workflows, QST providers can make quantum technologies accessible beyond quantum-expert developers, and align them more closely with real practitioner needs.

👉 **Explicitly communicate value through concrete and domain-specific examples.** Performance expectancy emerged as one of the strongest predictors of adoption intention. To reinforce perceived value, QST providers should complement high-level claims with concrete demonstrations of where quantum approaches offer measurable advantages. For example, Qiskit’s Quantum Machine Learning module¹² enables developers to prototype

¹⁰ <https://qiskit-community.github.io/qiskit-optimization/>

¹¹ <https://pennylane.ai/>

¹² <https://qiskit-community.github.io/qiskit-machine-learning/>

hybrid quantum–classical models—such as quantum kernels or variational quantum neural networks—using high-level abstractions that substantially reduce the need for low-level circuit design. Similarly, early industrial collaborations in areas like portfolio optimization illustrate how quantum-assisted solvers may provide improvements over classical techniques in selected optimization scenarios.¹³ Following principles of social persuasion (Cialdini et al. 2009), publicly reporting adoption trajectories, successful case studies, and clear evidence of performance improvements can further strengthen developers’ expectations and support more informed adoption decisions.

👉 **Promote social diffusion by increasing transparency and visibility.** Although social influence in our study affects intention more than actual usage behavior, community effects remain a key driver in the early adoption phases. To leverage this, quantum-software communities and platform developers should actively publish adoption metrics, highlight active contributors, and make visible the level and dynamics of open-source activity to foster network effects. For instance, widely used projects such as Qiskit and PennyLane already maintain public repositories where commit history, issue tracking, pull requests, contributor statistics, and community engagement are visible,¹⁴ allowing new developers to observe ecosystem vitality and community maturity before engaging. Such visibility can lower social and psychological barriers, increase trust, and encourage newcomers to adopt QSTs more confidently.

👉 **Support innovation-driven engagement.** Developers with high levels of personal innovativeness are more likely to adopt QSTs. Providers and community leaders can nurture this segment by organizing quantum hackathons, coding challenges, or open meetups that encourage hands-on experimentation. For example, initiatives such as the IBM Quantum Challenge¹⁵ or quantum machine-learning hackathons hosted by Xanadu and the PennyLane community¹⁶ provide structured opportunities for developers to engage with real QST platforms while solving domain-oriented problems. Such events attract early adopters, lower the practical barriers to trying quantum workflows, and foster sustained involvement within the ecosystem.

👉 **Strengthen internal knowledge-sharing routines.** Knowledge-sharing mechanisms such as brown-bag talks, team discussions, and pair programming can consolidate adoption internally, especially in organizations where QSTs are still unfamiliar or experimental. Concrete models for such initiatives already exist: for example, IBM Quantum promotes internal learning circles based on the Qiskit Textbook,¹⁷ while cloud providers such as AWS offer “Braket Immersion Days”¹⁸ to help teams jointly explore quantum workflows and hybrid development practices. Similarly, Microsoft provides workshop formats and training resources for teams adopting Azure Quantum.¹⁹ These structured knowledge-sharing routines help organizations build internal expertise, reduce uncertainty, and sustain engagement as QSTs evolve.

¹³ <https://www.ibm.com/quantum/blog/vanguard-portfolio-optimization>

¹⁴ <https://github.com/Qiskit>, <https://github.com/PennyLaneAI>

¹⁵ <https://challenges.quantum-computing.ibm.com>

¹⁶ <https://www.xanadu.ai/events>

¹⁷ <https://learning.quantum.ibm.com>

¹⁸ <https://aws.amazon.com/braket/>

¹⁹ <https://learn.microsoft.com/azure/quantum/>

👉 **Improve documentation and developer support.** Facilitating conditions were found to influence actual use of QSTs, highlighting the importance of accessible and reliable learning resources. Developers and platform providers should therefore prioritize accurate documentation, step-by-step tutorials, and clear API references. Existing initiatives already offer successful models: the Qiskit Textbook²⁰ provides structured explanations and executable examples, PennyLane’s hands-on tutorials²¹ support practical experimentation with variational and hybrid models, and AWS Braket maintains an extensive repository of ready-to-run examples²² for common workflows. Building on such approaches, QST platforms could further lower adoption barriers by integrating intelligent support features—such as AI-driven code recommendations, contextual help, or interactive debugging assistance—to guide developers through complex quantum programming tasks.

6.3.2 Research Implications and Future Directions

This study opens several avenues for future research and offers actionable recommendations for researchers investigating the socio-technical adoption of QSTs. The following points summarize key insights and opportunities for theory refinement, empirical investigation, and methodological innovation.

📖 **Refine and focus adoption models.** The validated model presented in this study can serve as a robust baseline for future investigations. Researchers may expand it by integrating additional constructs—such as perceived investment risk, switching costs, or long-term value expectations—or focus more narrowly on specific theoretical frameworks. In particular, constructs from the DoI theory demonstrated stronger explanatory power than those from UTAUT2. In addition, our work adopts a theory-based model precisely because it is the first in this research agenda; future studies should integrate our knowledge of more quantum-based constructs (e.g., quantum noise) to build more specific knowledge.

📖 **Account for individual and cultural variability in adoption.** Constructs such as compatibility, performance expectancy, or risk tolerance may be perceived differently depending on the background of the developer. Future work should consider developer personas (e.g., junior vs senior, academic vs industrial) as well as cultural dimensions such as uncertainty avoidance, to better understand heterogeneous adoption behavior.

📖 **Investigate the gap between intention and behavior.** While behavioral intention (BI) is a strong predictor of actual use (UB), certain constructs—such as social influence (SI)—affect intention without translating into usage. Future studies should explore the organizational and contextual barriers that hinder this transition, potentially through longitudinal or mixed-method research.

📖 **Study workplace mechanisms for habit formation.** As habit (Hb) emerged as the strongest predictor of actual use, further research is needed to investigate how organizational practices—such as onboarding processes, training programs, or project allocation strategies—can support sustained QST engagement over time.

📖 **Analyze the support ecosystem around QSTs.** Facilitating conditions (FC) were found to have a significant impact on usage. Researchers should assess the quality, ac-

²⁰ <https://qiskit.org/textbook>

²¹ <https://docs.pennylane.ai>

²² <https://github.com/aws/amazon-braket-examples>

cessibility, and effectiveness of documentation, tutorials, developer tools, and community platforms in reducing friction and supporting adoption.

Conduct qualitative research to identify friction points. Qualitative studies involving practitioners can reveal concrete barriers in the adoption journey—such as ambiguous documentation, steep learning curves, or unclear tool behavior—and translate them into actionable insights for both researchers and technology providers.

Conduct longitudinal and mixed-method studies in the future to validate the non-significant relationships. Building on the non-significant relationships observed in our model, future work should further investigate why constructs such as effort expectancy and perceived trust did not play a decisive role in QST adoption. This outcome may be influenced by the early-adopter profile of current QST practitioners and by the uniformly high complexity of available tools. Future studies—possibly longitudinal or mixed-method—should examine whether these constructs become more relevant as QST ecosystems mature and adoption expands beyond early adopters. Furthermore, as quantum hardware becomes more powerful and stable, and as concrete industrial use cases emerge, practitioners' adoption intentions may shift substantially. Longitudinal research is therefore essential to understand how perceptions, expectations, and adoption drivers evolve over time in response to these technological advancements.

6.4 Threats to Validity

As with any research, the findings presented in this study should be interpreted with awareness of certain limitations and threat to validity (Wohlin et al. 2012).

6.4.1 Conclusion Validity

Conclusion validity pertains to the reliability of the relationships identified between independent and dependent variables.

Potential threats in this category stem primarily from the statistical tests used in the analysis. To mitigate these risks, we relied on PLS-SEM, recognized for its robustness across diverse research scenarios. We strictly adhered to the methodological guidelines outlined by Hair Junior et al. (2014), ensuring methodological rigor. Furthermore, we utilized Smart-PLS, a widely recognized tool cited in over 1000 peer-reviewed studies (Hair Junior et al. 2014), to enhance the reliability of our findings.

We also examined local effect sizes (f^2) to assess the practical relevance of the structural paths. Following the guidelines of Hair Junior et al. (2014), values of 0.02, 0.15 and 0.35 represent small, medium, and large effects respectively. Our results show that Compatibility exerts the strongest influence in the model, with large effects on both PE ($f^2 = 0.839$) and EE ($f^2 = 0.515$). Inn shows medium effects on PE ($f^2 = 0.169$) and small-to-medium effects on EE ($f^2 = 0.144$). PE also has a small-to-medium effect on BI ($f^2 = 0.144$). Other paths, such as PV $\rightarrow$ BI ($f^2 = 0.058$), SI $\rightarrow$ BI ($f^2 = 0.033$), and BI $\rightarrow$ UB ($f^2 = 0.037$), exhibit small effects, whereas paths involving Trust, Facilitating Conditions, and Effort Expectancy display negligible impact. These results—together with the results on predictive power reported in Section 5.3.5—corroborate the interpretation of the model and support our results.

6.4.2 Internal Validity

Internal validity relates to the risk that factors influencing the dependent variables were not adequately considered.

To minimize such a risk, we grounded our study in established and well-validated theories within the domain of technology adoption. Moreover, we applied strict participant selection criteria to ensure that the sample accurately reflected the target population while maintaining heterogeneity, as supported by the demographic characteristics reported in Section 4.3.2.

A further potential threat to internal validity concerns endogeneity, which may arise from simultaneity or reverse causality among constructs such as PV, SI, or PE. For instance, developers who already intend to adopt QSTs may perceive the cost as more acceptable (affecting PV), actively seek reinforcing communities (affecting SI), or develop higher expectations regarding performance. To assess this risk, we applied the Gaussian Copula procedure, which is specifically designed to detect endogeneity in PLS-SEM models. As reported in Table X, none of the copula terms were statistically significant (e.g., $GC(PE \rightarrow BI) = 0.845, p = .398$; $GC(PV \rightarrow BI) = 1.458, p = .145$; $GC(SI \rightarrow BI) = 0.136, p = .891$), indicating no evidence of endogeneity. While these results support the exogeneity of our structural paths, we acknowledge that endogeneity remains an inherent concern in cross-sectional studies. Future research adopting longitudinal or experimental designs may further mitigate this risk.

An ulterior potential threat in our context concerns possible confounding factors—such as participants' prior experience with QSTs, their degree of exposure to quantum hardware, or familiarity with specific platforms—that could influence adoption-related perceptions. We mitigated these risks through a multi-stage screening procedure that ensured a variety of QST experience across participants. Additionally, core aspects of prior experience are inherently captured within the theoretical constructs employed in our model (e.g., innovativeness, compatibility, habit), which absorb part of the variability associated with different practitioner profiles. Furthermore, we employed PLS-SEM and all its tools in order to conduct multiple statistical analyses allowing the identification of problems concerning the omission of relevant variables. As reported in the results section, none of these cases occurred. Nonetheless, we acknowledge that—as in any study—it is impossible to construct a model capable of taking every possible factor into account. There is no doubt that our approach is methodologically sound for an initial study of this kind, but we encourage further studies with alternative constructs to expand knowledge on the adoption of QSTs and reinforce the validity of our study.

6.4.3 Construct Validity

Construct validity addresses the accuracy of the measurements used to capture the study variables.

All constructs in the research model were measured using validated instruments from prior studies (Venkatesh et al. 2012; Rogers 2003; Mun et al. 2006; Choi and Ji 2015; Nordhoff et al. 2021). The questionnaire design followed state-of-the-art survey methodology, incorporating best practices such as randomization of questions and attention-check

mechanisms to improve reliability (Kitchenham and Pfleeger 2008; Danilova et al. 2021). Moreover, we employed standardized instruments to evaluate the measurement model (e.g., Indicator Reliability and Internal Consistency Reliability tests); All tests (as discussed in Section 5.2) met the suggested criteria, confirming the reliability of the measurement.

6.4.4 External Validity

External validity concerns the extent to which the findings can be generalized beyond the study sample.

To enhance generalizability, we applied rigorous filtering criteria to the Prolific participant pool, ensuring that respondents met the necessary characteristics for our investigation. Additionally, we determined the required sample size using G*Power recommendations (Faul et al. 2009) to achieve statistical adequacy. However, generalizability remains a challenge, particularly given the early-stage nature of QSTs. As the technology matures, adoption patterns may evolve, potentially leading to different findings in future studies.

In addition, the characteristics of our sample strengthen its representativeness for the current population of QST practitioners. Participants display heterogeneous backgrounds in terms of professional roles, years of experience, and frequency of QST use, and our filtering strategy ensured that only individuals with verifiable exposure to QSTs were included. This diversity, combined with the quality controls applied during screening, provides a reasonable approximation of the developer population currently experimenting with or adopting quantum technologies.

Furthermore, our study focuses only on developers who have adopted QSTs at least to some extent during their careers; this choice excludes individuals who have never used the technology and therefore does not allow us to collect information on this category. We believe that this is a further opportunity to improve the work, capable of corroborating the data we have obtained and improving the generalisability of the study.

7 Conclusion

This study advances our understanding of QSTs adoption by providing the first theory-driven empirical examination of the mechanisms that influence developers' intentions and behaviours in this domain. By analysing the relative importance of established adoption determinants—many of which behave differently in QSTs compared to other technologies—the work clarifies which factors matter, which do not, and how these relationships manifest through direct, indirect, and total effects. This represents a foundational contribution in a field where assumptions have so far been largely conceptual. Moreover, the study meets its stated objectives by systematically modelling adoption determinants through a validated theoretical framework, empirically testing their relationships with a rigorous PLS-SEM design, and interpreting the findings in light of the concrete characteristics of QSTs. Together, these elements establish the baseline empirical knowledge required to inform more specialized quantum-specific adoption research going forward.

The findings highlight that compatibility with developers' workflows, personal innovativeness, and performance expectancy are the primary drivers of adoption. However, organizational support and usual exposure emerged as stronger determinants of actual use,

emphasizing the role of workplace interventions. Surprisingly, financial considerations negatively influenced adoption, and neither trust in technology nor effort expectancy played a significant role.

The key contributions of this study are the following:

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Author Contributions Both the authors, i.e., Stefano Lambiase and Andrea De Lucia, contributed to the study conception, design, and writing.

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Data Availability The authors declare that the data supporting the findings of this study are available within the paper and its supplementary information files available at the following link <https://figshare.com/s/57ed77894c88e8c1f2af>.

Declarations

Ethics approval Ethics approval was received from the Ethics Committee of University of Salerno, Italy.

Informed Consent Study participants received informed consent to participation and data were treated anonymously.

Conflict of Interest The authors declare no conflict of interest.

Clinical Trial Number Not applicable.

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